

Project Report- Irish Libraries

A report on a book reservation website and library system using PHP, MySQL, HTML, and CSS

Pages Used in My Application:

Common Header and Footer:

Every page on this website consists of a common navigation bar and footer.

Index.php

This is the website's homepage. It consists of a navigation bar at the top of the page, a welcome message accompanied by some links, and a footer. The layout slightly changes depending on the user's log in status. If they're not logged in, a log in and sign-up button are displayed underneath the welcome message and there's only one link in the navigation bar, "My Profile", which leads them to the log in page. If they are logged in, the two buttons are replaced with a message that uses PHP session variables, (`$_SESSION['username']`) to print the user's username to the screen. The "My Profile" link in the navigation bar also now redirects them to the actual My Profile page, and an option to log out is added. The user remains logged in until they choose to log out. Sessions are used throughout this project to allow the user to remain logged in and pass common session variables among the pages. The following 2 session variables are used throughout the project:

`$_SESSION['loggedIn']`

`$_SESSION['username']`

Interaction With Other Pages:

As Index.php is the home page, it doesn't interact much with other pages in this project, although it does provide links to redirect users to other pages (Signup.php and Login.php). It also uses sessions and session variables that are set in CheckLogin.php and Insert.php as previously mentioned.

Signup.php

This is the website's sign-up page to collect new user details and create an account. The PHP at the top pulls any error messages or form data out of the session (set by Insert.php), and unsets them to clear them after use. Then, a sign-up form is built using HTML, using those values so the user doesn't lose what they typed (eg. `echo htmlspecialchars($u)`) Each field has some PHP underneath that checks if there's an error message for that field and prints it.

Interaction With Other Pages:

When you click sign up, the form posts to Insert.php. If Insert.php finds problems, it sets `$_SESSION['errors']` and `$_SESSION['Data']`, then redirects back here. It also inserts the data sent by the form into the MySQL database if no errors are found. Like other pages, it also provides links to direct the user to Login.php and Index.php.

Insert.php

This file validates, cleans, and inserts the data into the MySQL database (the users table) after a user signs up. It's not a page the user can interact with, but it's very important. Firstly, it connects itself to the database through PHPConnectSQL.php. Then it validates all the inputs using a function that checks all mandatory fields are filled, username rules, password length and match, name and address formats, and mobile and telephone number format. If anything fails, it inserts an error message into an array \$errors. It then sets those error messages in the session in the else statement at the bottom and redirects back to Signup.php. If validation passes and \$errors is empty, it passes all the variables through another function cleanData, hashes the password with password_hash() and runs an INSERT query to add the new user to the users table. After that, it sets \$_SESSION['loggedIn'] and \$_SESSION['username'] so the user is automatically logged in. Finally, it redirects to Index.php.

NB – All data in this project is inserted into the database using prepare and execute statements as they protect against SQL injection.

Interaction With Other Pages:

Insert.php is only ever called by Signup.php's form. It either sends you back to Signup.php with errors or forward to Index.php if successful.

Login.php

This page is the login form. Like Signup.php, it pulls error messages and data from the session at the top (set by CheckLogin.php) and then unsets them. The form is simple and consists of username and password fields. It uses sessions to remember the username value, but not the password for security purposes. If there are errors (like empty fields or invalid login), they're shown underneath the inputs. These errors are handled by CheckLogin.php.

Interaction With Other Pages:

The form posts to CheckLogin.php. If that file finds errors, it sets \$_SESSION['loginErrors'] and \$_SESSION['Data'], then redirects back here. That's why Login.php has code to show those errors and keep the username value. If login succeeds, you're sent to Index.php. It also has links to direct you to Index.php or Signup.php.

CheckLogin.php

Like Insert.php, this file isn't a page the user can interact with, but it handles the log in logic. Its structure is similar to Insert.php. It connects to the database and uses a validation function to validate that the username and password aren't empty. If they are, it adds error messages to an array \$errors. It later sets these errors and sends the user back to Login.php. Otherwise, it prepares a SQL query to look up the user in the users table. As previously mentioned, a prepared statement was used with bind_param() to avoid SQL injection. If a row is found, it fetches the hashed password and checks it against the entered password using password_verify(). If it matches, it sets \$_SESSION['loggedIn'] and \$_SESSION['username'] and redirects to Index.php. If not, it sets an "invalid" error and sends you back to Login.php.

Interaction With Other Pages:

CheckLogin.php is only called by Login.php's form. It either sends you back to Login.php with errors or forward to Index.php if successful.

MyProfile.php

This is the most complex page. It can only be accessed by user's with an account. It consists of a search form to search books, and a section to view any books currently reserved by the user. The search form has 3 fields: Title and Author which can be typed, and Categories which is a dropdown select menu of all the categories in the database. The category options aren't hard-coded, they are retrieved from the database and displayed using a SELECT query and a while loop.

The program starts by connecting to the database and checking if the user is logged in — if not, it redirects to Login.php. Then it sets up variables for search filters (title, author, category) and pagination (page number, limit, offset). A flag, \$searchPerformed is used to check if the user has searched a book. If the user uses the form to search, i.e. \$searchPerformed === true, (the form is sent by GET), it builds a SQL query with LIKE conditions for title and author, and an exact match for category. This allows partial searching.

It also adds LIMIT ?, ? for pagination. The search query is prepared and executed, and results are displayed in a list. Each book shows title, author, category, ISBN, and either a "Reserve" button (this is a form posting to ReserveBook.php) or a message saying it's already reserved.

A count query is also performed to further aid pagination. It counts how many books match the user's search and stores the total amount in a variable \$totalRecords. The number of pages needed to display all results is then calculated by doing \$totalRecords/limit and using ceil() to round it up to the nearest whole number.

Pagination links are built at the bottom using the total record count. Separately, it always runs another query to show the user's reserved books, joining reservations, books, and categories. Each reserved book is listed with a "Cancel" button (form posting to CancelReservation.php).

Interaction With Other Pages:

MyProfile.php interacts with ReserveBook.php and CancelReservation.php through forms, and with the database through multiple queries. It's the page where searching, reserving, and cancelling all come together.

ReserveBook.php

This program relates to the reserve button from MyProfile.php. It starts by connecting to the database and checking the session to make sure the user is logged in. Then it gets the ISBN from \$_POST and the username from the session. It prepares an INSERT query into the reservations table with those values and the current date using NOW(). If the insert executes, it updates the "Reserved" column in books, changing the value to 1 (Reserved is Boolean). Then it redirects back to MyProfile.php.

Interaction With Other Pages:

ReserveBook.php is only called by MyProfile.php's reserve form. Like Insert.php and CheckLogin.php, it doesn't show its own page — it just processes the request and sends you back. It also has links to direct you back to home or log out.

CancelReservation.php

This program has a similar structure to ReserveBook.php but does the opposite task . Like reserving a book, it connects to the database, checks the session, and gets the ISBN and username. However, instead of preparing an INSERT query, it prepares a DELETE query from the reservations table. If the query executes, it updates the "Reserved" column in the books table, setting the value to 0. Then it redirects back to MyProfile.php.

Interaction With Other Pages:

CancelReservation.php is only called by MyProfile.php's cancel form. Like ReserveBook.php, it doesn't show its own page — it just processes and redirects.

Logout.php

This file logs the user out. It starts the session, clears \$_SESSION, calls session_destroy(), and then redirects to Index.php. It's very short but important.

Interaction With Other Pages:

Logout.php is linked from Index.php and MyProfile.php. It doesn't show a page — it just ends the session and sends you home.

PHPConnectSQL.php

This file sets up the database connection. It defines the server name, username, password, and database name, then creates a new mysqli connection. If the connection fails, it calls die() with an error message. When building this project, I had it print a message "Connection successful" if the connection succeeded. This was needed for testing purposes and handling errors, but was removed when the project was completed.

Interaction With Other Pages:

Every program that needs the database (Insert.php, CheckLogin.php, MyProfile.php, ReserveBook.php, CancelReservation.php) includes this file with require_once. That way, they all share the same connection setup.

Design Document

System Architecture

This project was built using a mix of PHP, MySQL, HTML, and CSS. PHP handles all the server-side logic like logging in, signing up, searching, and reserving books. MySQL is the database where all the user accounts, books, categories, and reservations are stored. HTML and CSS are used to build the pages and style them, so they look consistent.

Structure:

- Frontend: Pages like Index.php, Signup.php, Login.php, and MyProfile.php are mostly HTML with embedded PHP. They display forms, results, and navigation. They provide an interface for the user to interact with.
- Backend scripts: Files like Insert.php, CheckLogin.php, ReserveBook.php, and CancelReservation.php don't show their own pages. They just process form submissions, update the database, and redirect back to the right page.
- Database connection: All the scripts that need the database include PHPConnctSQL.php, which sets up the \$conn object. This keeps the connection code in one place.
-

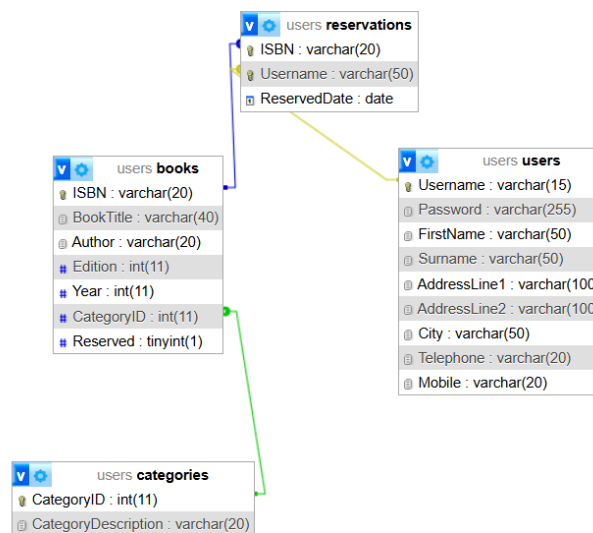
Database Design

The database has four main tables:

- **users** - Stores account details. Fields include Username (required), Password (required), FirstName (required), Surname (required), AddressLine1 (required), AddressLine2, City, Telephone, and Mobile (required). This table is used by Insert.php when creating accounts and CheckLogin.php when logging in. As instructed, each row is uniquely identified by username, making it the table's primary key.
- **books** - Stores information about each book. Fields include ISBN, BookTitle, Author, Edition, Year, CategoryID, and Reserved. This table is used by MyProfile.php when searching for books. It's also used by ReserveBook.php and CancelReservation.php when updating the "Reserved" column. ISBN is the table's primary key, CategoryID is a foreign key, linking the books table to the categories table.
- **categories** Stores the list of categories/genres of books. Fields include CategoryID and CategoryDescription. This table is used by MyProfile.php for the category dropdown and filter searches. CategoryID is this table's primary key.
- **reservations** Stores which user has reserved which book. Fields include ISBN, Username, and ReservedDate. This table is used by ReserveBook.php when making a reservation, CancelReservation.php when cancelling, and MyProfile.php when showing reserved books. It has a composite primary key – ISBN and Username, which are also foreign keys linking it to the books and users table.

The Relationships:

- Each book belongs to a category (books.CategoryID → categories.CategoryID).
- Each reservation links a book to a user (reservations.ISBN → books.ISBN, reservations.Username → users.Username).

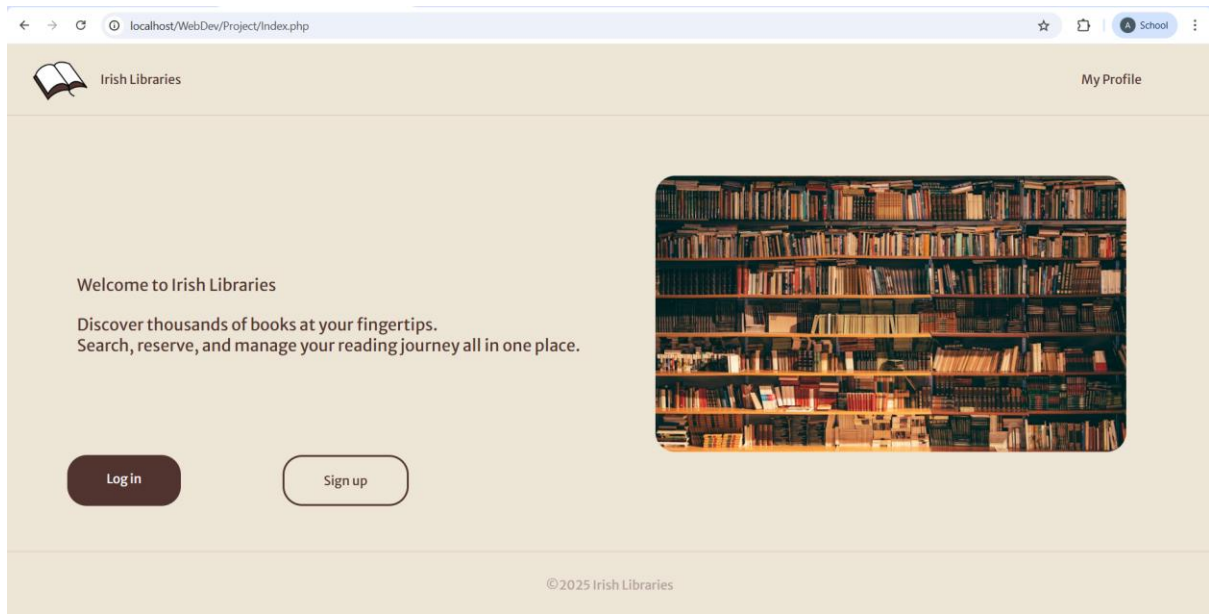


Design Choices

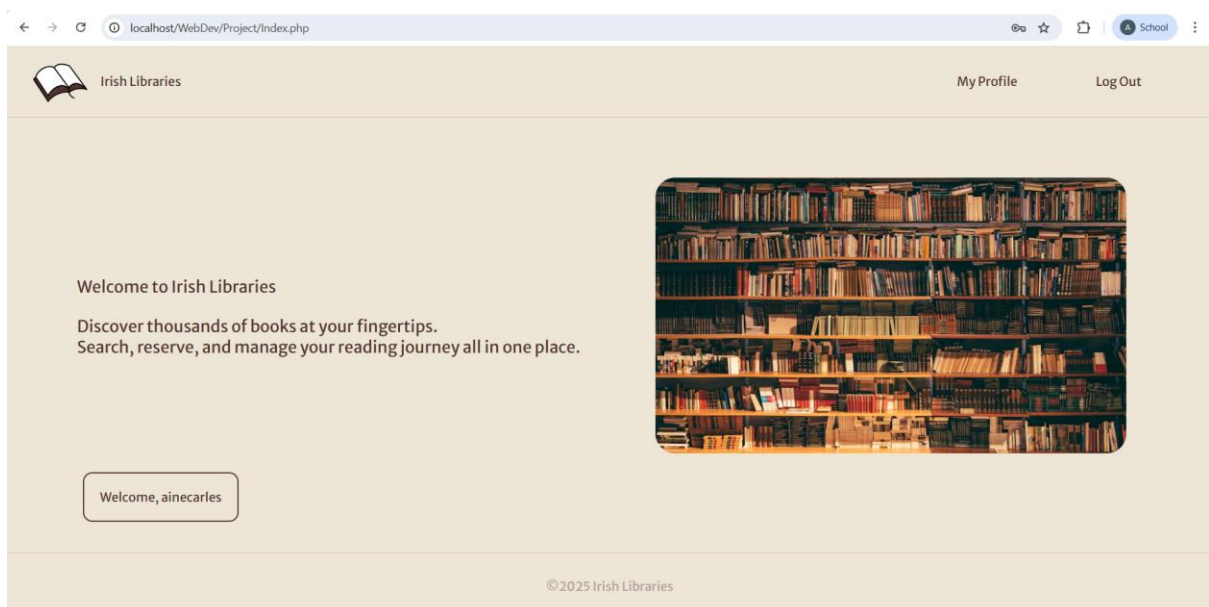
- Sessions: Used to track whether a user is logged in and who they are. This made it easy to personalise pages and manage access.
- Prepared statements: All database queries use prepared statements with `bind_param()`. This prevents SQL injection and makes the code safer.
- Password hashing: Passwords are stored using `password_hash()` and checked with `password_verify()`. This is more secure than storing plain text.
- Sticky forms: Signup and login forms remember what you typed if there are errors. This improves usability and makes the site feel more polished.
- htmlspecialchars: `htmlspecialchars()` was used throughout the project to safely display user input and prevent HTML or script injection.
- Pagination: Search results are limited to five per page, with links to move between pages. This keeps the results tidy and avoids huge lists.
- Consistent styling: CSS was used for colours, spacing, and fonts. This keeps the look consistent across all pages.

Copy Samples of Each Screen in my Application

Index.php – Not Logged In



Index.php – Logged In



Signup.php (Allows for scrolling but I zoomed out the page so you can see all of it)

Irish Libraries

Home Log In

Sign Up

Create your account to start reserving books

Username *
(E.g. user_123)

Password *
(Min. 8 characters)

Confirm Password *
(Re-enter password)

First Name *
(John)

Surname *
(Doe)

Address Line 1 *
(E.g. 10 Main Street)

Address Line 2
(Apartment, suite etc. (optional))

City
(Dublin)

Telephone
(E.g. 0123456789 (optional))

Mobile *
(E.g. 087 8617890 (optional))

Sign Up

[Already have an account? Log In here](#)

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Login.php

Irish Libraries

Home Sign Up

Log In

Username

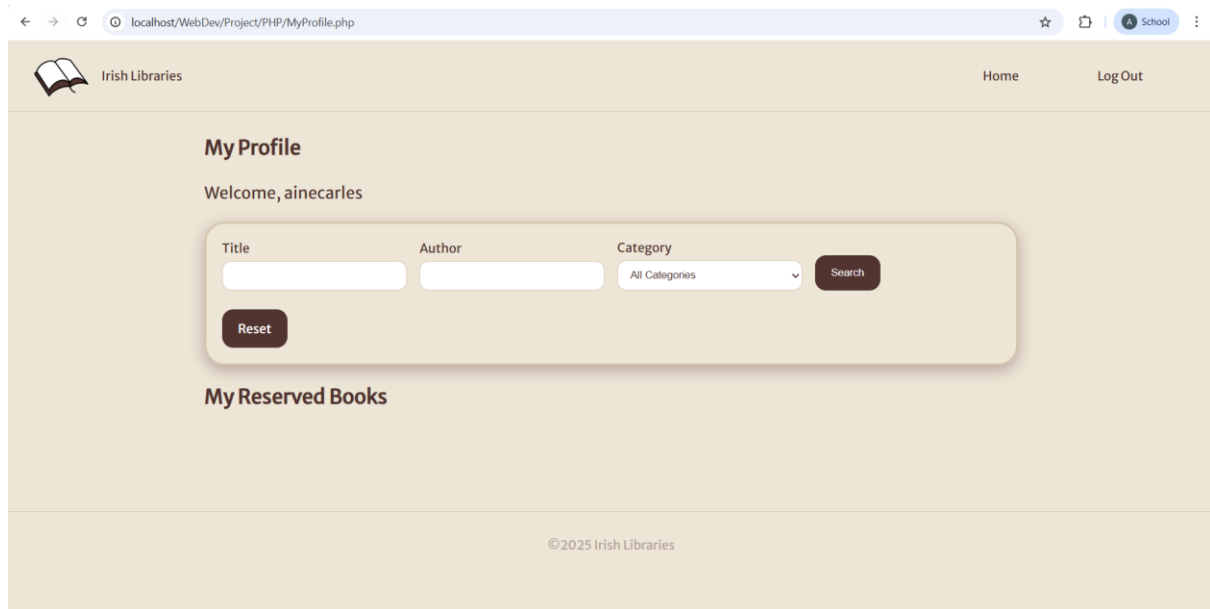
Password

Log In

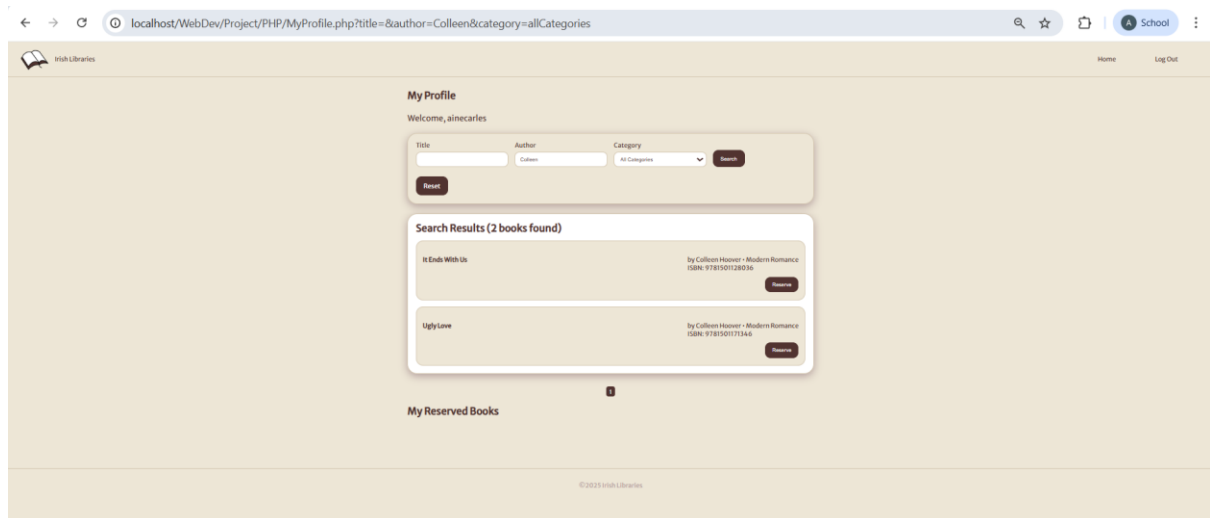
[Don't have an account? Sign up here](#)

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MyProfile.php – Before Searching



MyProfile.php – After Searching (Allows for scrolling but I zoomed out the page so you can see all of it)



MyProfile.php – After Reserving

← → ↻ 🌐 localhost/WebDev/Project/PHP/MyProfile.php 🔍 ☆ 📄 School ⋮

 Irish Libraries Home Log Out

My Profile

Welcome, ainecarles

Title

Author

Category

All Categories

Search

Reset

My Reserved Books

- **It Ends With Us**
by Colleen Hoover • Modern Romance
ISBN: 9781501128036

Cancel

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Copy Listings of My Source Code:

Index.php

```
<?php
session_start();

$loggedIn = $_SESSION['loggedIn'] ?? false;
$username = $_SESSION['username'] ?? "";
?>

<!DOCTYPE html>
<html lang="en">
<!-- Boiler Plate -->

<head>
    <meta charset="UTF-8" />
    <title>Irish Libraries</title>
    <meta name="viewport" content="width=device-width,initial-scale=1" />
    <meta name="description" content="Library site to search, reserve, and view all the books a user
has reserved" />
    <link rel="stylesheet" href="CSS/Index.css">
</head>

<body>

    <!-- Nav Bar-->
    <div class="container">

        <div class="navbar">
```

```
<div class="logo">

</div>
```

```
<p class="navTitle">Irish Libraries</p>
```

```
<ul>

    <li><!-- Display Different Options on NavBar if User is Logged In-->

        <?php if(!empty($_SESSION['loggedIn']) && $_SESSION['loggedIn'] === true):?>

            <li><a href="PHP/MyProfile.php">My Profile</a></li>

            <li><a href="PHP/Logout.php">Log Out</a></li>

        <?php else: ?>

            <a href="PHP/login.php">My Profile</a>

        <?php endif; ?>

    </li>

</ul>

</div>

</div>
```

```
<div class="pageContent">
```

```
<div class="leftContent">
```

```
<p class="welcome"> Welcome to Irish Libraries</p>
```

```
<p>Discover thousands of books at your fingertips. <br> Search, reserve, and manage your
reading journey all in one place.<p>
```

```
</div>
```

```
<?php if(!empty($_SESSION['loggedIn']) && $_SESSION['loggedIn'] === true):?>
```

```
<div class="lm">
```

```
<p class="ll">Welcome, <?php echo htmlspecialchars($username);?></p> <!-- Welcome
Message if User is Logged In-->
```

```
</div>
```

```
<?php else: ?>
```

```
<div class="buttonRow">
```

```
<a href="PHP/Login.php" class="btn1">Log in</a>
```

```
<a href="PHP/Signup.php" class="btn2">Sign up</a>
```

```
</div>
```

```
<?php endif; ?>
```

```
<div class="rightImg">
```

```

```

```
</div>
```

```
</div>
```

```
<footer>©2025 Irish Libraries</footer>
```

```
</body>
```

```
</html>
```

Signup.php

```
<?php
```

```
session_start();
```

```
$errors = $_SESSION['errors'] ?? [];
```

```
$formData = $_SESSION['formData'] ?? [];
```

```
$u = $formData['username'] ?? "";
```

```
$f = $formData['firstName'] ?? "";
```

```
$s = $formData['surname'] ?? ";  
$a1 = $formData['addressLine1'] ?? ";  
$a2 = $formData['addressLine2'] ?? ";  
$c = $formData['city'] ?? ";  
$t = $formData['telephone'] ?? ";  
$m = $formData['mobile'] ?? ";  
  
unset($_SESSION['errors'], $_SESSION['formData']); //Clear after use  
?>
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<!-- Boiler Plate -->
```

```
<head>
```

```
    <meta charset="UTF-8" />
```

```
    <title>Irish Libraries Signup</title>
```

```
    <meta name="viewport" content="width=device-width,initial-scale=1" />
```

```
    <meta name="description" content="Sign Up Page" />
```

```
    <link rel="stylesheet" href="../CSS/Signup.css">
```

```
</head>
```

```
<body>
```

```
    <!-- Nav Bar -->
```

```
    <div class="container">
```

```
        <div class="navbar">
```

```
            <div class="logo">
```

```
                
```

</div>

<p class="navTitle">Irish Libraries</p>

Home

Log In

</div>

</div>

<div class="pageContent">

<h3 class="header">Sign Up</h3>

<p class="header">Create your account to start reserving books</p>

<!-- Sign up form -->

<div class="formBox">

<form action="Insert.php" method="post">

<div class="field">

<label for="username" class="required">Username</label>

<input type="text" name="username" placeholder="Eg: user_123"

value="<?php echo htmlspecialchars(\$u); ?>"><!--Remember input values -->

<!-- Display Username Errors -->

<?php

if (isset(\$errors['emptyU'])) echo "<p class='error'>{\$errors['emptyU']}</p>";

if (isset(\$errors['whiteSpaceU'])) echo "<p class='error'>{\$errors['whiteSpaceU']}</p>";

if (isset(\$errors['incorrectCharU'])) echo "<p
class='error'>{\$errors['incorrectCharU']}</p>";

if (isset(\$errors['tooLongU'])) echo "<p class='error'>{\$errors['tooLongU']}</p>";

```

        if (isset($errors['duplicateU'])) echo "<p class='error'>{$errors['duplicateU']}</p>";
    ?>
</div>

<div class="twoCol">
    <div class="field">
        <label for="password" class="required">Password</label>
        <input type="password" name="password" placeholder="Min. 6 characters">

        <!-- Display Password errors -->

        <?php
            if (isset($errors['emptyP'])) echo "<p class='error'>{$errors['emptyP']}</p>";
            if (isset($errors['noMatchP'])) echo "<p class='error'>{$errors['noMatchP']}</p>";
            if (isset($errors['shortP'])) echo "<p class='error'>{$errors['shortP']}</p>";
        ?>
    </div>

    <div class="field">
        <label for="confirmPassword" class="required">Confirm Password</label>
        <input type="password" name="confirmPassword" placeholder="Re-enter password">

        <?php if (isset($errors['emptyCP'])) echo "<p
class='error'>{$errors['emptyCP']}</p>";?>
    </div>
</div><!-- End twoCol -->

<div class="twoCol">
    <div class="field">
        <label for="firstName" class="required">First Name</label>
        <input type="text" name="firstName" placeholder="John"
            value="<?php echo htmlspecialchars($f); ?>">

```



```
<!-- Display First Name Errors -->

<?php
    if (isset($errors['emptyFN'])) echo "<p class='error'>{$errors['emptyFN']}</p>";

    if (isset($errors['incorrectCharFN'])) echo "<p
class='error'>{$errors['incorrectCharFN']}</p>";

    ?>

</div>
```

```
<div class="field">

    <label for="surname" class="required">Surname</label>

    <input type="text" name="surname" placeholder="Doe"
value="<?php echo htmlspecialchars($s); ?>">
```

```
<!-- Display Surname Errors -->

<?php
    if (isset($errors['emptyS'])) echo "<p class='error'>{$errors['emptyS']}</p>";

    if (isset($errors['incorrectCharS'])) echo "<p
class='error'>{$errors['incorrectCharS']}</p>";

    ?>

</div>
```

```
</div><!-- End Two Col -->
```

```
<div class="field">

    <label for="addressLine1" class="required">Address Line 1</label>

    <input type="text" name="addressLine1" placeholder="Eg. 24 Main Street"
value="<?php echo htmlspecialchars($a1); ?>">
```

```
<!-- Display Address Errors -->
```

```
<?php
    if (isset($errors['emptyA1'])) echo "<p class='error'>{$errors['emptyA1']}</p>";
```

```
        if (isset($errors['incorrectCharA1'])) echo "<p  
class='error'>{$errors['incorrectCharA1']}</p>";
```

```
    ?>
```

```
</div>
```

```
<div class="field">
```

```
    <label for="addressLine2">Address Line 2</label>
```

```
    <input type="text" name="addressLine2" placeholder="Apartment, suite etc. (optional)"
```

```
    value="<?php echo htmlspecialchars($a2);?>">
```

```
    <!-- Display Address Errors -->
```

```
    <?php
```

```
        if (isset($errors['incorrectCharA2'])) echo "<p  
class='error'>{$errors['incorrectCharA2']}</p>";
```

```
    ?>
```

```
</div>
```

```
<div class="field">
```

```
    <label for="city">City</label>
```

```
    <input type="text" name="city" placeholder="Dublin"
```

```
    value="<?php echo htmlspecialchars($c); ?>">
```

```
    <!-- Display City Errors -->
```

```
    <?php
```

```
        if (isset($errors['incorrectCharC'])) echo "<p  
class='error'>{$errors['incorrectCharC']}</p>";
```

```
    ?>
```

```
</div>
```

```
<div class="twoCol">
```

```
    <div class="field">
```

```
        <label for="telephone">Telephone</label>
```

```
<input type="tel" name="telephone" placeholder="Eg. 019296673 (optional)"
value="<?php echo htmlspecialchars($t); ?>">
```

```
<!-- Display Telephone Errors -->
```

```
<?php
```

```
if (isset($errors['errorT'])) echo "<p class='error'>{$errors['errorT']}</p>";
```

```
?>
```

```
</div>
```

```
<div class="field">
```

```
<label for="mobile" class="required">Mobile</label>
```

```
<input type="tel" name="mobile" placeholder="Eg. 0873647689 (10 digits)"
```

```
value="<?php echo htmlspecialchars($m); ?>">
```

```
<!-- Display Mobile Phone Errors -->
```

```
<?php
```

```
if (isset($errors['emptyM'])) echo "<p class='error'>{$errors['emptyM']}</p>";
```

```
if (isset($errors['errorM'])) echo "<p class='error'>{$errors['errorM']}</p>";
```

```
?>
```

```
</div>
```

```
</div><!-- End Two Col -->
```

```
<div>
```

```
<input type="submit" value="Sign Up">
```

```
</div>
```

```
<!-- Redirect to Login Page-->
```

```
<div class="redirect">
```

```
<p> Already have an account? <a href="Login.php"> Log in here </a></p>
```

```
</div>
```

```
</form>
```

```
</div><!-- End formBox -->
```

```
<footer>©2025 Irish Libraries</footer>
```

```
</div><!-- End pageContent -->
```

```
</body>
```

```
</html>
```

Insert.php

```
<?php
```

```
session_start();
```

```
//Connect to the database
```

```
require_once "PHPConnnectSQL.php";
```

```
//Errors array to store errors
```

```
$errors = [];
```

```
//Function for Validation
```

```
function validation($u, $p, $cp, $f, $s, $a1, $a2, $c, $t, $m, &$errors, $conn){
```

```
    //Username:
```

```
    if(empty($u)){
```

```
        $errors['emptyU'] = "Username must be filled";
```

```
    }
```

```
    else if(preg_match("/\s/", $u)){
```

```

    $errors['whiteSpaceU'] = "Username can't have spaces";
}

else if(!preg_match("/^[A-Za-z0-9_]+$/", $u)){
    $errors['incorrectCharU'] = "Username can only contain, letters, numbers, and underscore";
}

else if(strlen($u) > 15){
    $errors['tooLongU'] = "Username has a maximum of 15 characters";
}

else{//Check for duplicate usernames in database
    $stmt = $conn->prepare("SELECT Username FROM users WHERE Username = ?");
    $stmt->bind_param("s", $u);
    $stmt->execute();
    $stmt->store_result();

    if($stmt->num_rows > 0){
        $errors['duplicateU'] = "Username already taken";
    }

    $stmt->close();
}

//Password
if(empty($p)){
    $errors['emptyP'] = "Password must be filled";
}

else if($p !== $cp){

```

```
$errors['noMatchP'] = "Passwords don't match";
}

else if(strlen($p) < 6){
    $errors['shortP'] = "Password must be at least 6 characters long";
}

//Confirm Password
if(empty($cp)){
    $errors['emptyCP'] = "Confirm Password must be filled";
}

//First Name
if(empty($f)){
    $errors['emptyFN'] = "First name must be filled";
}

else if(!preg_match("/^[A-Za-z'-]+$/", $f)){
    $errors['incorrectCharFN'] = "Names can only contain letters and the following: ' -";
}

//Surname
if(empty($s)){
    $errors['emptyS'] = "Surname must be filled";
}

else if(!preg_match("/^[A-Za-z'-]+$/", $s)){
    $errors['incorrectCharS'] = "Names can only contain letters and the following: ' -";
}
```

```

//Address Line 1
if(empty($a1)){
    $errors['emptyA1'] = "Address Line 1 must be filled";
}

else if(!preg_match("/^[A-Za-z0-9 ,.'-]+$/", $a1)){
    $errors['incorrectCharA1'] = "Addresses can only contain numbers, letters and the following: , . '
-";
}

//Address Line 2
if(!empty($a2)){
    if(!preg_match("/^[A-Za-z0-9 ,.'-]+$/", $a2)){
        $errors['incorrectCharA2'] = "Addresses can only contain numbers, letters and the following: ,
. ' -";
    }
}

//City
if(!empty($c)){
    if(!preg_match("/^[A-Za-z ,.'-]*$/", $c)){
        $errors['incorrectCharC'] = "City can only contain letters and the following: , . ' -";
    }
}

//Telephone
if(!empty($t)){
    if(!preg_match("/^(0[124569]\d{5,7})$/", $t)){//Telephone is 7-9 digits long
        $errors['errorT'] = "Invalid starting digits telephone must be at least 7 digits";
    }
}

```

```
}
```

```
//Mobile
```

```
if(empty($m)){
```

```
    $errors['emptyM'] = "Mobile must be filled";
```

```
}
```

```
else if(!preg_match("/^08\d{8}$/", $m)){
```

```
    $errors['errorM'] = "Mobile number must start with 08 and be 10 digits";
```

```
}
```

```
}//End validation
```

```
//Function for cleaning input
```

```
function cleanData($data){
```

```
    $data = trim($data);
```

```
    $data = stripslashes($data);
```

```
    $data = htmlspecialchars($data);
```

```
    return $data;
```

```
}//End cleanData
```

```
if($_SERVER['REQUEST_METHOD'] === 'POST'){
```

```
    $u = $_POST['username'];
```

```
    $p = $_POST['password'];
```

```
    $cp= $_POST['confirmPassword'];
```

```
    $f = $_POST['firstName'];
```

```
    $s = $_POST['surname'];
```

```
    $a1 = $_POST['addressLine1'];
```

```
    $a2 = $_POST['addressLine2'];
```

```
    $c = $_POST['city'];
```



```
$t = $_POST['telephone'];
```

```
$t = preg_replace("/^D/", "", $t); //Replace everything that isn't a digit with an empty string
```

```
$m = $_POST['mobile'];
```

```
$m = preg_replace("/^D/", "", $m);;
```

```
validation($u, $p, $cp, $f, $s, $a1, $a2, $c, $t, $m, $errors, $conn);
```

```
if(empty($errors)){
```

```
    //Clean data
```

```
    $u = cleanData($u);
```

```
    $f = cleanData($f);
```

```
    $s = cleanData($s);
```

```
    $a1 = cleanData($a1);
```

```
    $a2 = cleanData($a2);
```

```
    $c = cleanData($c);
```

```
    $t = cleanData($t);
```

```
    $m = cleanData($m);
```

```
    //Hash password for security
```

```
    $hashedPwd = password_hash($p, PASSWORD_DEFAULT);
```

```
    //Insert data into database
```

```
    $stmt = $conn->prepare("INSERT INTO users (Username, Password, Firstname, Surname,  
AddressLine1, AddressLine2, City, Telephone, Mobile)
```

```
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)"); // $stmt = statement object to hold SQL, ? =  
placeholder
```

```
    $stmt->bind_param("ssssssss",
```

```
$u,  
$hashedPwd,  
$f,  
$s,  
$a1,  
$a2,  
$c,  
$t,  
$m); //Bind the value. "s" - Type of parameter, $u etc -Inject this into the ?
```

```
if($stmt->execute()){ //Then execute it
```

```
    $_SESSION['loggedIn'] = true;
```

```
    $_SESSION['username'] = $u;
```

```
    $stmt->close();
```

```
    $conn->close();
```

```
    header("Location: ../Index.php");
```

```
    exit;
```

```
} //End if
```

```
} //End if
```

```
else{
```

```
    $conn->close();
```

```
    $_SESSION['errors'] = $errors;
```

```
    $_SESSION['formData'] = $_POST;
```

```
    header("Location: Signup.php");
```

```
    exit;
```

```
} //End Else
```

```
} //End if
```

?>

Login.php

<?php

//Set and Unset Session Variables

session_start();

\$errors = \$_SESSION['loginErrors'] ?? "";

\$Data = \$_SESSION['Data'] ?? [];

\$user = \$Data['username'] ?? "";

unset(\$_SESSION['loginErrors'], \$_SESSION['Data']);

?>

<!DOCTYPE html>

<html lang="en">

<!-- Boiler Plate -->

<head>

<meta charset="UTF-8" />

<title>Irish Libraries Login</title>

<meta name="viewport" content="width=device-width,initial-scale=1" />

<meta name="description" content="Sign Up Page" />

<link rel="stylesheet" href="../CSS/Login.css">

</head>

<body>

<!-- Nav Bar -->

<div class="container">

```
<div class="navbar">
```

```
    <div class="logo">
```

```
        
```

```
    </div>
```

```
    <p class="navTitle">Irish Libraries</p>
```

```
    <ul>
```

```
        <li><a href="../Index.php">Home</a></li>
```

```
        <li><a href="Signup.php">Sign Up</a></li>
```

```
    </ul>
```

```
</div>
```

```
</div>
```

```
<div class="pageContent">
```

```
    <h3>Log In</h3>
```

```
    <div class="formBox">
```

```
        <form action="CheckLogin.php" method="post">
```

```
            <div class="field">
```

```
                <label for="username">Username</label>
```

```
                <input type="text" name="username"
```

```
                    value="<?php echo htmlspecialchars($user);?>"><!-- Remember Input Values -->
```

```
            <!-- Display username error -->
```

```
                <?php if(isset($errors['emptyUser'])) echo "<p  
class='error'>{$errors['emptyUser']}</p>";?>
```

```
            </div> <!-- End field -->
```

```

<div class="field">

    <label for="password">Password</label>

    <input type="password" name="password">


    <!-- Display password error -->

    <?php if(isset($errors['emptyPass'])) echo "<p
class='error'>{$errors['emptyPass']}</p>";?>

</div> <!-- End field -->


<!-- Error: If username or password not found -->

<?php if(isset($errors['invalid'])) echo "<p class='error'>{$errors['invalid']}</p>";?>


<div>

    <input type="submit" value="Log In">

</div>


<div class="redirect">

    <p>Don't have an account? <a href="Signup.php">Sign up here</a></p>

</div><!-- End redirect -->

</form>

</div> <!-- End formBox -->


<footer>©2025 Irish Libraries</footer>


</div><!-- End pageContent -->

</body>

</html>

```

CheckLogin.php

```

<?php
session_start();

```

```
//Connect to Database

require_once "PHPConnctSQL.php";


//Errors array to store errors

$errors = [];


//Function To Catch and Display Errors

function validation($user, $pass, &$errors) {

    if (empty($user)) {
        $errors['emptyUser'] = "Username must be filled";
    }

    if (empty($pass)) {
        $errors['emptyPass'] = "Password must be filled";
    }
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {

    $user = trim($_POST['username']);
    $pass = $_POST['password'];

    validation($user, $pass, $errors);

    //If basic errors

    if (!empty($errors)) {
        $_SESSION['loginErrors'] = $errors;
        $_SESSION['Data'] = $_POST;
        header("Location: Login.php");
    }
}
```

```

        exit;
    }//End if

    //If not check database
    $stmt = $conn->prepare("SELECT Username, Password FROM users WHERE Username = ?");
    $stmt->bind_param("s", $user);
    $stmt->execute();
    $result = $stmt->get_result();

    //If username found
    if ($result->num_rows == 1) {

        $row = $result->fetch_assoc();//Fetch the matching row

        //Check password
        if (password_verify($pass, $row['Password'])) { //password_verify - Verifies that the given hash
matches the given password

            //Successful Login
            $_SESSION['loggedIn'] = true;
            $_SESSION['username'] = $row['Username'];

            header("Location: ../Index.php");
            exit;
        }//End if
    }//End if

    //If username not found OR password wrong
    $errors['invalid'] = "Incorrect username or password";
    $_SESSION['loginErrors'] = $errors;
    $_SESSION['Data'] = $_POST;

```

```
header("Location: Login.php");

exit;

}

?>
```

MyProfile.php

```
<?php

session_start();

require_once "PHPConnctSQL.php";

//If User isn't Logged in, Redirect to Login Page
if (!isset($_SESSION['loggedIn']) || !$_SESSION['loggedIn']) {
    header("Location: Login.php");
    exit;
}

//Set Session Variable
$username = $_SESSION['username'] ?? "";

// Function to Clean Data
function cleanData($data) {
    $data = trim($data);
    $data = stripslashes($data);
    return $data;
}

$title = cleanData($_GET['title'] ?? "");
$author = cleanData($_GET['author'] ?? "");
$category = $_GET['category'] ?? 'allCategories';
```



```

// Pagination Variables (only used if a search happens)

$page = isset($_GET['page']) ? (int)$_GET['page'] : 1; //If the URL includes page, use it as the page
number. Otherwise, set page to 1.

if ($page < 1) { $page = 1; }

$limit = 5;

$startFrom = ($page - 1) * $limit;


// Flag to check if the user performed a search

$searchPerformed = ($_SERVER['REQUEST_METHOD'] === 'GET' && !empty($_GET)); //Returns
a boolean


// If a search was performed, run the search query

if ($searchPerformed) {
    $sql = "SELECT b.*, c.CategoryDescription
            FROM books b
            JOIN categories c ON b.CategoryID = c.CategoryID
            WHERE 1"; //Append AND easily

    $params = []; //Array to Store Variables to be Injected
    $types = ""; //String to store data types

    if (!empty($title)) {
        $sql .= " AND b.BookTitle LIKE ?";
        $params[] = "%$title%";
        $types .= "s";
    }

    if (!empty($author)) {
        $sql .= " AND b.Author LIKE ?";
        $params[] = "%$author%";
        $types .= "s";
    }
}

```

```

if (!empty($category) && $category !== "allCategories") {
    $sql .= " AND c.CategoryDescription = ?";
    $params[] = $category;
    $types .= "s";
}

$sql .= " LIMIT ?, ?"; // Append limit command to end of og query
$params[] = $startFrom; // add startFrom and limit to params
$params[] = $limit;
$types .= "ii";

$stmt = $conn->prepare($sql);
if (!empty($params)) {
    $stmt->bind_param($types, ...$params);
}
$stmt->execute();
$results = $stmt->get_result();

// Count total for pagination
$countSql = "SELECT COUNT(*) AS total
            FROM books b
            JOIN categories c ON b.CategoryID = c.CategoryID
            WHERE 1";

$countParams = [];
$countTypes = "";

if (!empty($title)) {
    $countSql .= " AND b.BookTitle LIKE ?";
    $countParams[] = "%$title%";
    $countTypes .= "s";
}

```

```

if (!empty($author)) {
    $countSql .= " AND b.Author LIKE ?";
    $countParams[] = "%$author%";
    $countTypes  .= "s";
}

if (!empty($category) && $category !== "allCategories") {
    $countSql .= " AND c.CategoryDescription = ?";
    $countParams[] = $category;
    $countTypes  .= "s";
}

$countStmt = $conn->prepare($countSql);

if (!empty($countParams)) {
    $countStmt->bind_param($countTypes, ...$countParams);
}

$countStmt->execute();

$countResult = $countStmt->get_result();//Get result of the count query

$totalRecords = (int)$countResult->fetch_assoc()['total'];//Get the result of the count query as an
associative array("total" -> result)

$totalPages  = max(1, (int)ceil($totalRecords / $limit));//ceil() - rounds up to nearest whole number
}

// Always load reserved books (Doesnt depend on user making a search)
$reserveSql = "SELECT b.ISBN, b.BookTitle, b.Author, r.ReservedDate, c.CategoryDescription
FROM reservations r
JOIN books b ON r.ISBN = b.ISBN
JOIN categories c ON b.CategoryID = c.CategoryID
WHERE r.Username = ?";

```

```
$reserveStmt = $conn->prepare($reserveSql);  
$reserveStmt->bind_param("s", $username);  
$reserveStmt->execute();  
$reserveResult = $reserveStmt->get_result();  
?>
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<!-- Boiler Plate -->
```

```
<head>
```

```
    <meta charset="UTF-8" />
```

```
    <title>Irish Libraries</title>
```

```
    <meta name="viewport" content="width=device-width,initial-scale=1" />
```

```
    <meta name="description" content="Library site to search, reserve, and view all the books a user  
has reserved" />
```

```
    <link rel="stylesheet" href="../CSS/MyProfile.css">
```

```
</head>
```

```
<body>
```

```
    <!-- Nav Bar-->
```

```
    <div class="container">
```

```
        <div class="navbar">
```

```
            <div class="logo">
```

```
                
```

```
            </div>
```

```
            <p class="navTitle">Irish Libraries</p>
```

```
<ul>

    <li><a href="../Index.php"> Home</a></li>

    <li><a href="Logout.php">Log Out</a></li>

</ul>

</div>

</div>
```

```
<div class="pageContent">
```

```
<h3>My Profile</h3>
```

```
<p>Welcome, <?php echo htmlspecialchars($username);?></p><!-- Welcome Message Displays
Username-->
```

```
<div class="searchBox">
```

```
<form class="searchForm" action="<?php echo
htmlspecialchars($_SERVER["PHP_SELF"]);?>" method="get">
```

```
<div class="field">
```

```
<label for="title">Title</label>
```

```
<input type="text" name="title" value="<?= htmlspecialchars($title) ?>"><!-- = is
shorthand for echo-->
```

```
</div><!-- End field -->
```

```
<div class="field">
```

```
<label for="author">Author</label>
```

```
<input type="text" name="author" value="<?= htmlspecialchars($author) ?>">
```

```
</div><!-- End field -->
```

```
<div class="field">
```

```
<label for="category">Category</label>
```

```
<select name="category">
```

```
        <option value="allCategories" <?=( $category === "allCategories") ? 'selected' : "
?>>All Categories</option><!-- Have allCategories as default when page first loads -->
```

```
<!-- Retrieve Values from Category Table -->
```

```
<?php
```

```
$sql = "SELECT CategoryDescription FROM categories";
```

```
$result = $conn->query($sql);
```

```
while ($row = $result->fetch_assoc()) {
```

```
    $cat = htmlspecialchars($row['CategoryDescription']);
```

```
    $selected = ($category === $row['CategoryDescription']) ? 'selected' : ""; //Is chosen
category ($category) = this row's category?
```

```
    echo "<option value=\"\$cat\" $selected>$cat</option>";
```

```
}
```

```
$conn->close();
```

```
?>
```

```
</select>
```

```
</div><!-- End field -->
```

```
<div class="searchActions">
```

```
    <button type="submit" class="btn">Search</button>
```

```
</div>
```

```
</form><!-- End Form -->
```

```
<div class="searchActions">
```

```
    <a href="MyProfile.php" class="btn">Reset</a>
```

```
</div>
```

```
</div><!-- End searchBox -->
```

```
<!-- Display Search Results Below Form -->
```

```
<?php if (isset($results) && $results->num_rows > 0): ?>
```

```
<div class="searchResults">
```

```
    <h4>Search Results (<?=$totalRecords ?> books found)</h4>
```

```
    <ul>
```

```
        <?php while ($row = $results->fetch_assoc()): ?>
```

```
            <li>
```

```
                <strong><?= htmlspecialchars($row['BookTitle']) ?></strong><br>
```

```
                by <?= htmlspecialchars($row['Author']) ?> • <?=
htmlspecialchars($row['CategoryDescription']) ?><br>
```

```
                ISBN: <?= htmlspecialchars($row['ISBN']) ?><br>
```

```
                <?php if ($row['Reserved']): ?>
```

```
                    <span class="unavailable">Already Reserved</span>
```

```
                <?php else: ?>
```

```
                    <form method="post" action="ReserveBook.php">
```

```
                        <input type="hidden" name="isbn" value="<?=
htmlspecialchars($row['ISBN']) ?>">
```

```
                        <button type="submit">Reserve</button>
```

```
                    </form>
```

```
                <?php endif; ?>
```

```
            </li>
```

```
        <?php endwhile; ?>
```

```
    </ul>
```

```
</div>
```

```
<?php
```

```
// Build Links for Pagination
```

```
echo '<ul class="pagination">';
```

```
for ($i = 1; $i <= $totalPages; $i++) {
```

```
    if($i == $page){
```

```

    $active = "class='active'";
}
else{
    $active = "";
}

```

echo "<li \$active>urlencode() - Ensures strings are safe to insert into a URL

```

"&author=" . urlencode($author) .
"&category=" . urlencode($category) . ">$i</a></li>";
}
echo '</ul>';
?>

```

<?php elseif (\$searchPerformed): ?><!-- First if statement = false, but search was still performed so this runs-->

```

<p>No books found matching your criteria.</p>
<?php endif; ?>

```

<!-- View Reserved Books -->

<h4>My Reserved Books</h4>

<?php while(\$row = \$reserveResult->fetch_assoc()):?>

<?= htmlspecialchars(\$row['BookTitle']) ?>

by <?= htmlspecialchars(\$row['Author']) ?> • <?= htmlspecialchars(\$row['CategoryDescription']) ?>

ISBN: <?= htmlspecialchars(\$row['ISBN']) ?>

<form action="CancelReservation.php" method="post">

<input type="hidden" name="isbn" value="<?php echo \$row['ISBN'];?>">

<button type="submit" class="cancelBtn">Cancel</button>


```
<?php endwhile; ?>
```

```
</div><!-- End pageContent -->
```

```
<footer>©2025 Irish Libraries</footer>
```

```
</body>
```

```
</html>
```

ReserveBook.php

```
<?php
```

```
session_start();
```

```
require_once "PHPConntectSQL.php";
```

```
//Ensure User is Logged In
```

```
if(!isset($_SESSION['username'])) {
```

```
    header("Location: Login.php");
```

```
    exit;
```

```
}
```

```
if($_SERVER['REQUEST_METHOD'] === 'POST'){
```

```
    $isbn = $_POST['isbn'];
```

```
    $username = $_SESSION['username'];
```

```
//Insert Reservation
```

```
$sql = "INSERT INTO reservations (ISBN, Username, ReservedDate)
```

```
    VALUES (?, ?, NOW());
```

```
$stmt = $conn->prepare($sql);
```

```
$stmt->bind_param("ss", $isbn, $username);
```

```

if($stmt->execute()){
    //Update Reserved in books
    $updateSql = "UPDATE books SET Reserved = 1 WHERE ISBN = ?";
    $updateStmt = $conn->prepare($updateSql);
    $updateStmt->bind_param("s", $isbn);
    $updateStmt->execute();
}

}

// Redirect back to profile
header("Location: MyProfile.php");
exit;

?>

```

CancelReservation.php

```

<?php
session_start();

require_once "PHPConnctSQL.php";

//Redirect to Login Page if User isn't logged in
if (!isset($_SESSION['username'])) {
    header("Location: Login.php");
    exit;
}

if($_SERVER['REQUEST_METHOD'] === 'POST'){

    $isbn = $_POST['isbn'];

```

```
$username = $_SESSION['username'];
```

```
//Delete reservation
```

```
$sql = "DELETE FROM reservations WHERE ISBN = ? AND Username = ?";
```

```
$stmt = $conn->prepare($sql);
```

```
$stmt->bind_param("ss", $isbn, $username);
```

```
if($stmt->execute()){
```

```
    //Reset Reserved in books after deletion
```

```
    $updateSql = "UPDATE books SET Reserved = 0 WHERE ISBN = ?";
```

```
    $updateStmt = $conn->prepare($updateSql);
```

```
    $updateStmt->bind_param("s", $isbn);
```

```
    $updateStmt->execute();
```

```
}
```

```
header("Location: MyProfile.php");
```

```
exit;
```

```
}
```

```
?>
```

Logout.php

```
<?php
```

```
session_start();
```

```
// Unset all session variables
```

```
$_SESSION = array();
```

```
// Destroy the session
```

```
session_destroy();
```

```
// Redirect to home page
```

```
header("Location: ../Index.php");
```

```
exit;
```

```
?>
```

PHPConnctSQL.php

```
<?php
```

```
$servername = "localhost";
```

```
$username = "root";
```

```
$password = "";
```

```
$dbname = "users";
```

```
//Create connection
```

```
$conn = new mysqli($servername, $username, $password, $dbname);
```

```
//Check connection
```

```
if($conn -> connect_error){
```

```
    die("Connection failed: ". $conn -> connect_error);
```

```
}
```

```
?>
```

